

# Qatar-1b

R-2764

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_150915 · created 2026-09-03T00:44:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457280.6605 +/- 0.00062 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.191 +/- 0.055                  |
| Transit depth (Rp/Rs)^2                           | 3.64 +/- 2.12 [%]                |
| Orbital Inclination (inc)                         | 88.92 +/- 2.87                   |
| Airmass coefficient 1 (a1)                        | 0.095 +/- 0.07                   |
| Airmass coefficient 2 (a2)                        | 1.87 +/- 1.7                     |
| Scatter in the residuals of the lightcurve fit is | 87.31 %                          |
| Best Comparison Star                              | None                             |
| Optimal Aperture                                  | 3.45                             |
| Optimal Annulus                                   | 11.29                            |
| Transit Duration (day)                            | 0.0829 +/- 0.0068                |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.191 ± 0.055 vs 0.1463 (deviation 0.71σ) |
| Duration fit vs published | 0.0829 d vs 0.0692 d (deviation 1.78σ)    |
| Mid-time O–C              | -0.9 min                                  |
| Depth SNR                 | 3.1                                       |
| Residual scatter          | 87.31 %                                   |

## Light curve

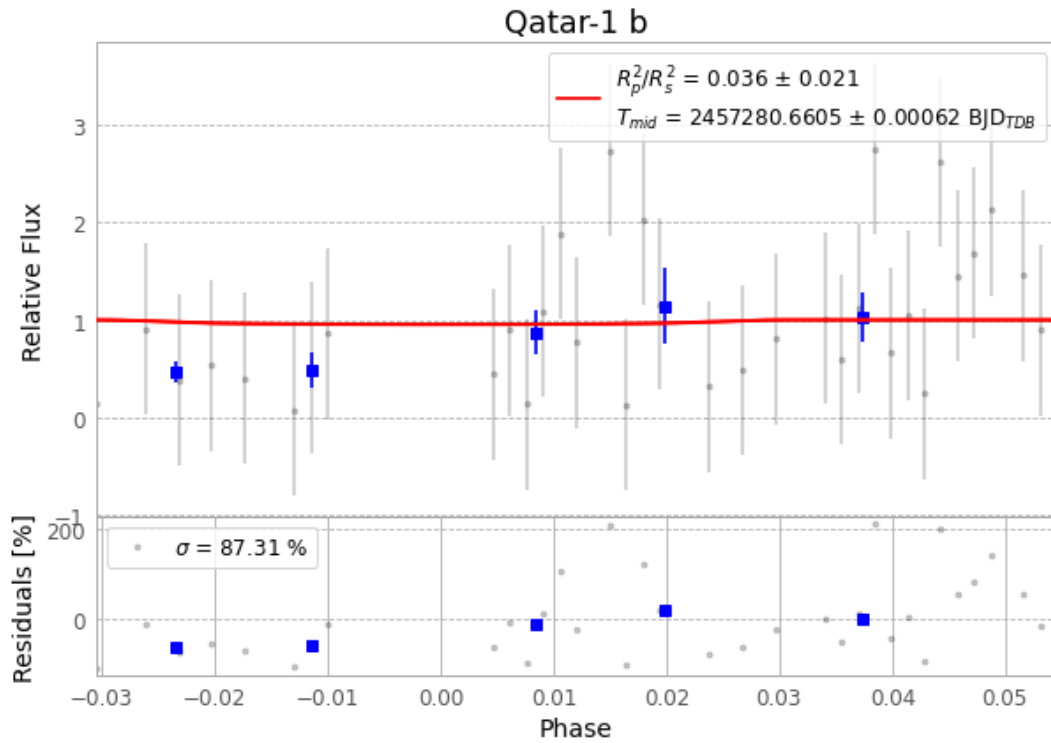


Figure 1 — FinalLightCurve\_Qatar-1 b\_2015-09-14.png (SHA-256 6ef0d1a3917ef3f7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [41, 400], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 f2d1ce0c3fa34533...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

60 FITS frames (39 MB), Qatar-1150915024213.FITS → Qatar-1150915053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-150915.log (ecd7331be93d...), FinalLightCurve\_Qatar-1 b\_2015-09-14.png (6ef0d1a3917e...), FinalLightCurve\_Qatar-1 b\_2015-09-14.csv (7692066e89cb...)

## Compute resources

Wall time 253.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:44Z.